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Software Engineering

Design patterns & Modern Architecture

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Agenda

- Introduction
  - SOLID Principles
- Design Patterns
  - Selected Patterns
- From Patterns to Architecture
- Clean Code & DDD
- Modern Architectural Approaches

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Introduction

Motivation

Why pure Object-Orientetion is not enough

Typical symptoms of naive OOP projects:

- **Tight coupling** between classes.
- So-called **"God objects"** knowing and doing too much.
- **Mixed concerns**: Business logic tangled with infrastructure (user interface, database, communication).

As a result we end up with **code that works, but resists change**.

- How to deal with that?

Foundations

SOLID Principles

Before discussing patterns, we must understand the principles of good object-oriented design.

- **S – Single Responsibility Principle**: A class should have one, and only one, reason to change. One task, one purpuse, etc.
- **O – Open/Closed Principle**: Entities should be open for extension, but closed for modification.
- **L – Liskov Substitution Principle**: Subtypes must be substitutable for their base types.
- **I – Interface Segregation Principle**: Many client-specific interfaces are better than one general-purpose interface. Client should not be forced to implement unnecssary methods that they don't use.
- **D – Dependency Inversion Principle**: Depend upon abstractions, not concrete concepts.

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Design Patterns

Design Patterns

What are they?

Design Pattern: Definition

A design pattern is a **proven and reusable solution** to a **recurring software design problem** that:

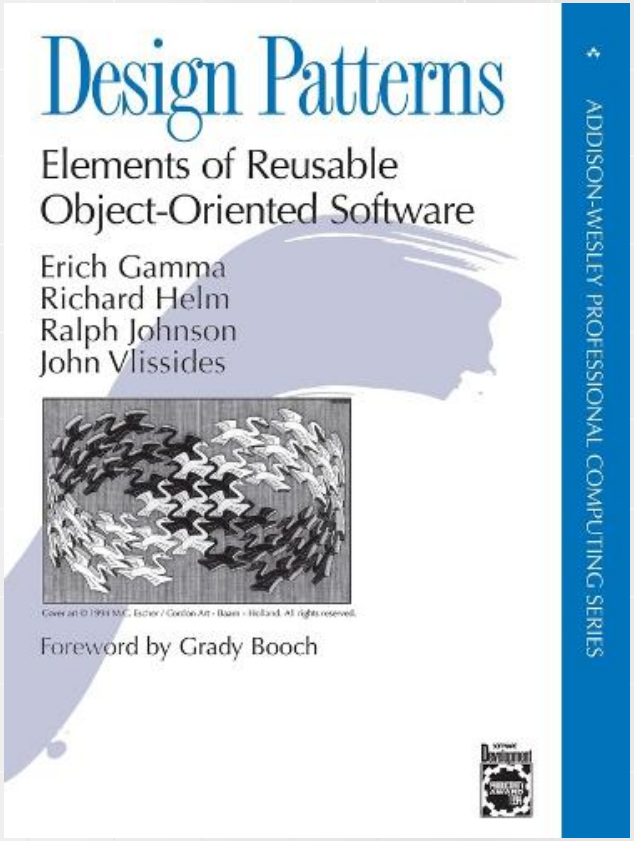
- applies in a **specific design context**
- describes a **general solution**, not a concrete implementation
- captures **best practices** derived from experience
- provides a **shared vocabulary** for design communication
- focuses on **structure and interaction** of software components

- Not ready-made solutions (copy-paste code).
- Not a library of classes.
- **A shared language** for communicating design decisions.
- Proven solutions to common problems in a specific context.

Famous Gang of Four (Gamma et al.)

- What is this famous GoF (Gamma, Helm, Johnson, Vlissides)?

Erich Gamma et al. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Reading, Massachusetts: Addison-Wesley. ISBN: 0-201-63361-2



Design Patterns

Classification (GoF)

- **Creational**
  - Control object creation
  - Examples: Singleton, Factory
- **Structural**
  - Organize class/object composition
  - Examples: Adapter, Decorator
- **Behavioral**
  - Assign responsibilities and communication
  - Examples: Observer, Strategy

Selected Patterns

Singleton

Intent: Ensure that a class has **only one instance** and provide a **global access point** to that instance.

Problem: Some objects (e.g. configuration managers, loggers) must be unique. Uncontrolled creation of instances may lead to **inconsistent state** or **resource conflicts**.

Solution (Idea): Restrict object creation by:

- making the constructor **private**,
- storing a **single static instance**,
- exposing a **static access method**.

Consequences

- ✓ Controlled access to a single shared instance
- ✓ Lazy initialization (optional)
- × Introduces **global state** (hidden dependencies)
- × Makes **testing and mocking harder**
- × Often violates the **Single Responsibility Principle**

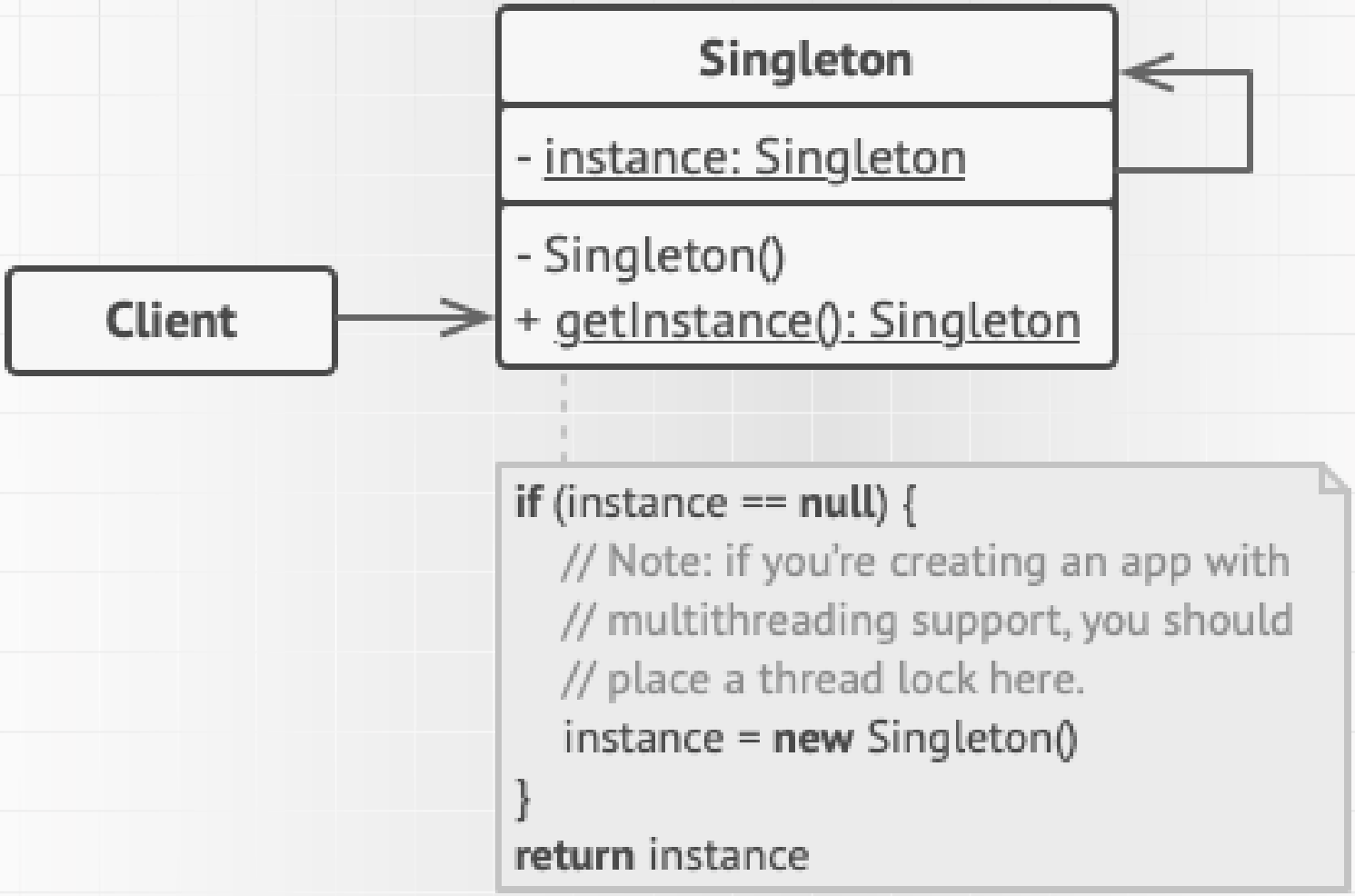


Figure: Source: refactoring.guru

- Avoid singletons, prefer **dependency injection** (you will see what is it in a few slides).

Selected Patterns

Factory / Factory Method

Intent: Define an interface for creating objects, while allowing subclasses to decide which concrete class to instantiate.

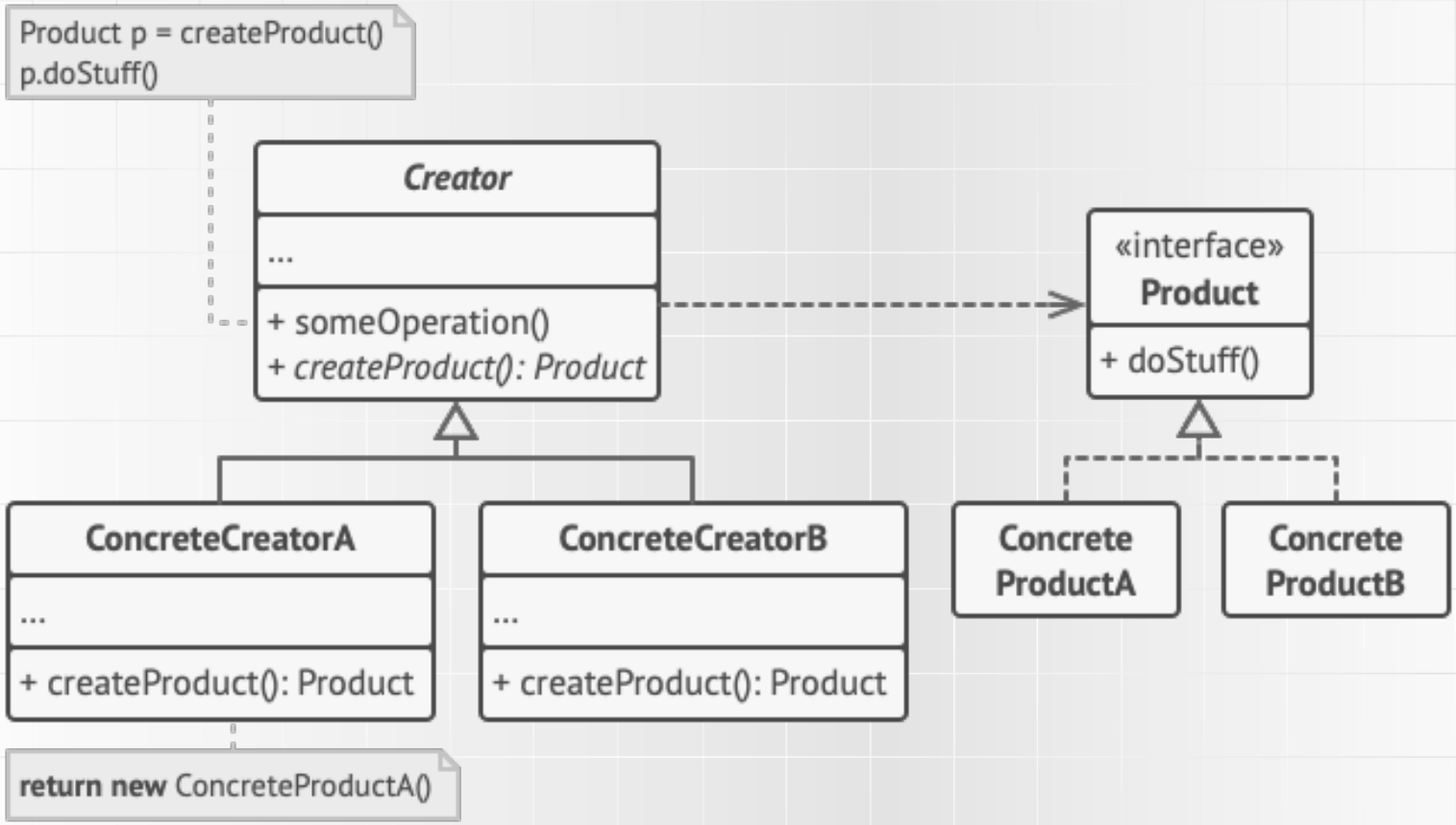
Problem: Direct object creation using new tightly couples the client to concrete classes, making the system **harder to extend** and **maintain**.

Solution: Encapsulate object creation by:

- defining a **factory interface or method**,
- delegating the choice of the **concrete class** to subclasses,
- returning objects through a **common abstraction**.

Consequences

- ✓ Separates **creation logic** from **usage logic**
- ✓ Supports the **Open/Closed Principle**
- ✓ Improves **extensibility** and **testability**
- × Introduces **additional classes** and indirection
- × May increase overall **code complexity**



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Selected Patterns

Adapter

Intent: Convert the interface of an existing class into another interface expected by the client, allowing incompatible classes to work together.

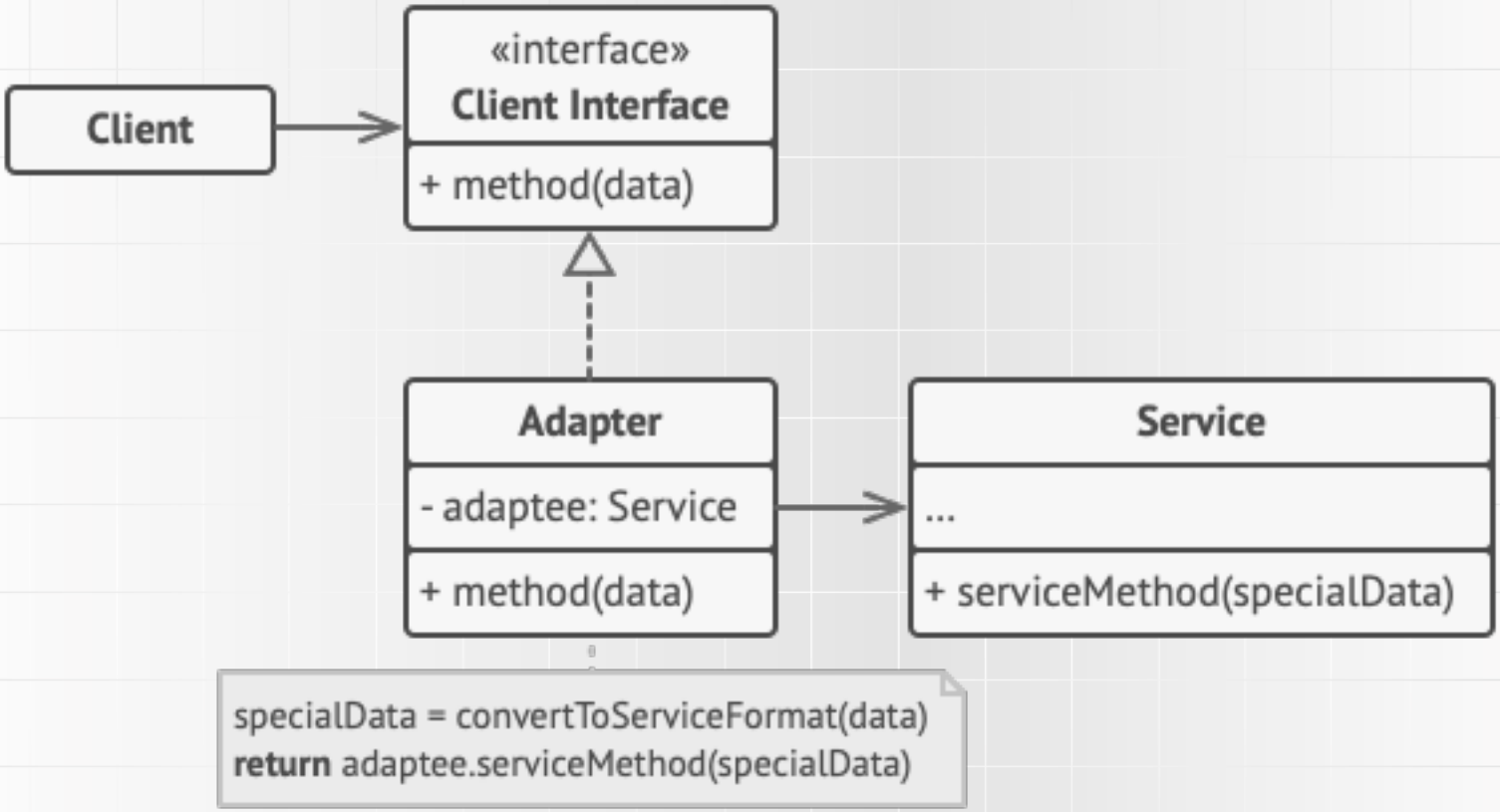
Problem: Existing (often legacy or third-party) classes expose interfaces that are **incompatible** with the client code, and cannot be modified directly.

Solution: Introduce an adapter that:

- implements the **target interface** expected by the client,
- internally wraps the **adaptee** (existing class),
- translates requests from the client into calls understood by the adaptee.

Consequences

- ✓ Enables integration of **legacy** and **third-party** components
- ✓ Decouples the client from concrete implementations
- ✓ Preserves the **Single Responsibility Principle**
- × Adds an extra layer of **indirection**
- × May increase the number of classes in the system



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Selected Patterns

Decorator

Intent: Attach additional responsibilities to an object dynamically. Decorators provide a flexible alternative to subclassing for extending behavior.

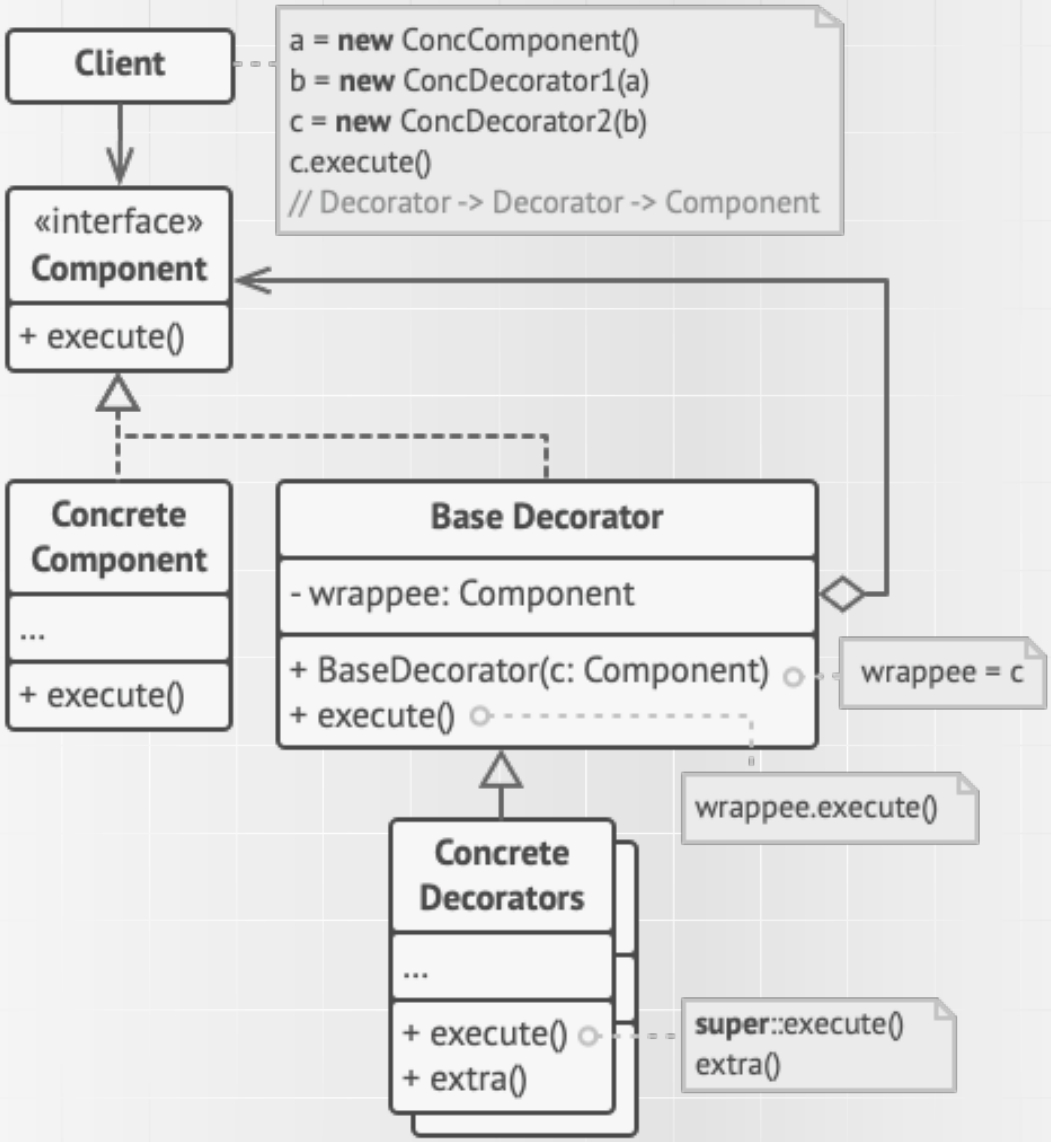
Problem: Extending behavior using inheritance leads to a **class explosion** and rigid designs, especially when behavior combinations are required.

Solution: Wrap objects using decorators that:

- implement the same **component interface**,
- hold a reference to a **wrapped object**,
- add behavior **before and/or after** delegating calls.

Consequences

- ✓ Supports the **Open/Closed Principle**
- ✓ Enables **dynamic composition** of behavior
- ✓ Favors **composition over inheritance**
- × Introduces additional **objects and indirection**
- × Can make the system **harder to understand and debug**



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- **Decorator** changes behavior and Adapter changes the interface.

Selected Patterns

Observer

Intent: Define a one-to-many dependency between objects so that when one object changes state, all its dependents are notified and updated automatically.

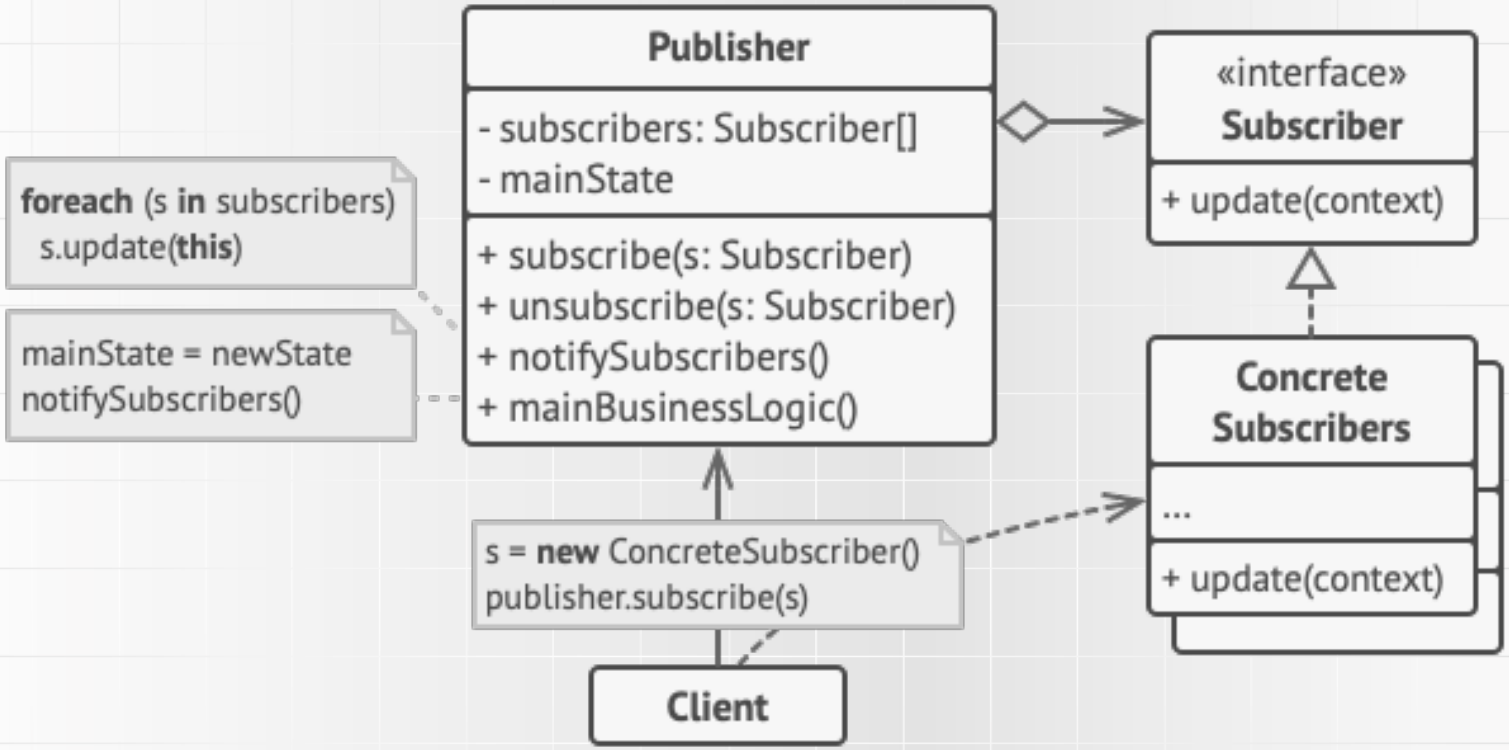
Problem: Multiple objects depend on the state of another object, but tight coupling would make the system **hard to extend** and **maintain**.

Solution: Introduce a subject that:

- maintains a list of **observers**,
- provides methods to **attach** and **detach** observers,
- automatically **notifies** observers when its state changes.

Consequences

- ✓ Enables **reactive and event-driven** behavior
- ✓ Promotes **loose coupling** between subject and observers
- ✓ Supports the **Open/Closed Principle**
- × May lead to **memory leaks** (Lapsed Listener Problem)
- × Can cause **unexpected update chains**



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Selected Patterns

Strategy

Intent: Define a family of algorithms, encapsulate each one, and make them interchangeable so that the algorithm can vary independently from its clients.

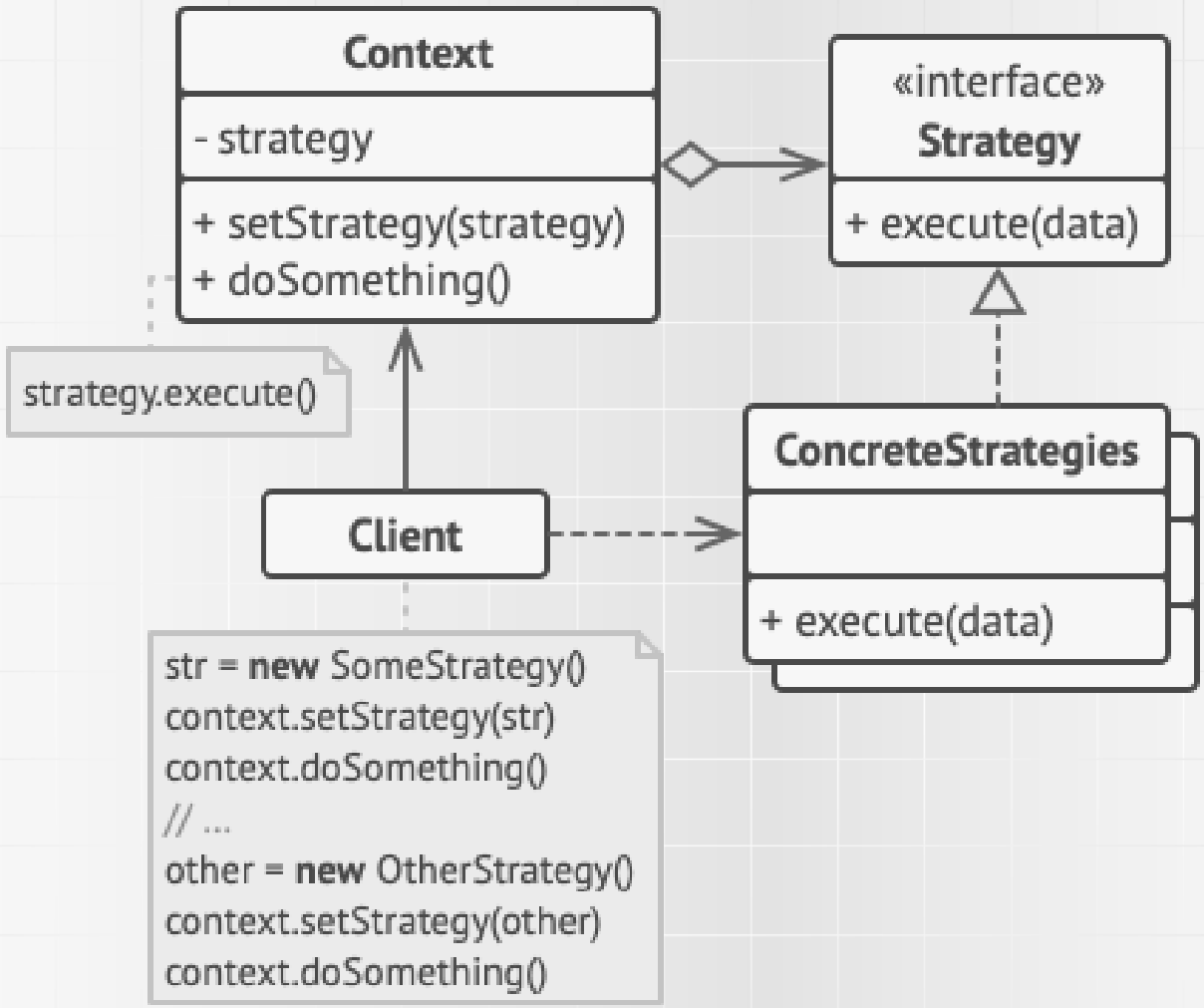
Problem: Multiple algorithmic variants implemented using conditional logic (`if/else`, `switch`) lead to **rigid** and **hard-to-maintain** code.

Solution: Encapsulate algorithms by:

- defining a common **strategy interface**,
- implementing each algorithm in a **separate class**,
- delegating algorithm selection to the **context**.

Consequences

- ✓ Eliminates complex conditional statements
- ✓ Allows **runtime switching** of algorithms
- ✓ Supports the **Open/Closed Principle**
- ✓ Improves **testability** of algorithms
- × Increases the number of **classes** in the system



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From Patterns to Architecture

Limits of Design Patterns

- GoF OOP design patterns improve quality **locally**, but they do not solve architectural problems.

Patterns work at the level of classes and objects. They do not address:

- Overall system structure.
- Layer dependencies.
- Module responsibilities.

Design vs Architecture

Design Pattern	**Architectural Pattern**
Local problem	Global decision
Classes / Objects	Modules / Layers / Components
Implementation detail	High-level structure
e.g., Strategy, Factory	e.g., MVC, Microservices, Hexagonal

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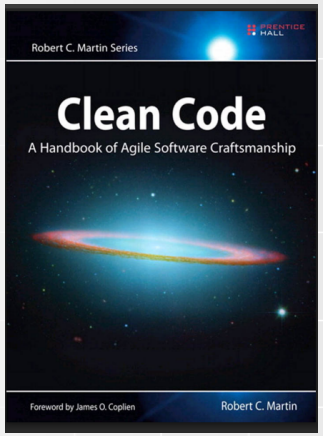
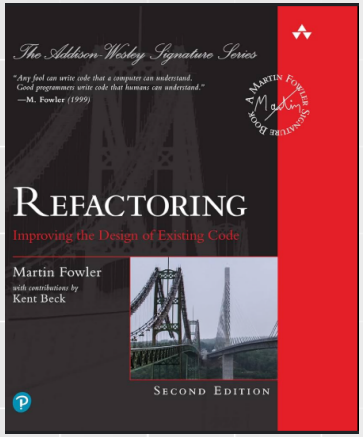
Clean Code & DDD

Clean Code – Prerequisite for good architecture

- Good architecture requires good code.





- Code easy to read and comprehensible
- Easy to change
- Tests, naming convention, exceptions, formatting, comments
- „code smells” → bad code heuristics (M.Fowler)
  - long method, god object, duplication, primitive obsession, shotgun surgery

Domain-Driven Design (DDD)

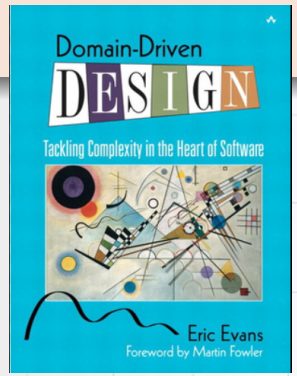
Core Concepts

Domain-Driven Design (DDD)

Domain-Driven Design is an approach to software development that focuses on modeling the **core business domain** and expressing it explicitly in the code.

DDD emphasizes:

- a **rich domain model** that reflects business concepts,
- a **shared language** between developers and domain experts (Ubiquitous Language),
- clear **boundaries** separating the domain from technical concerns.



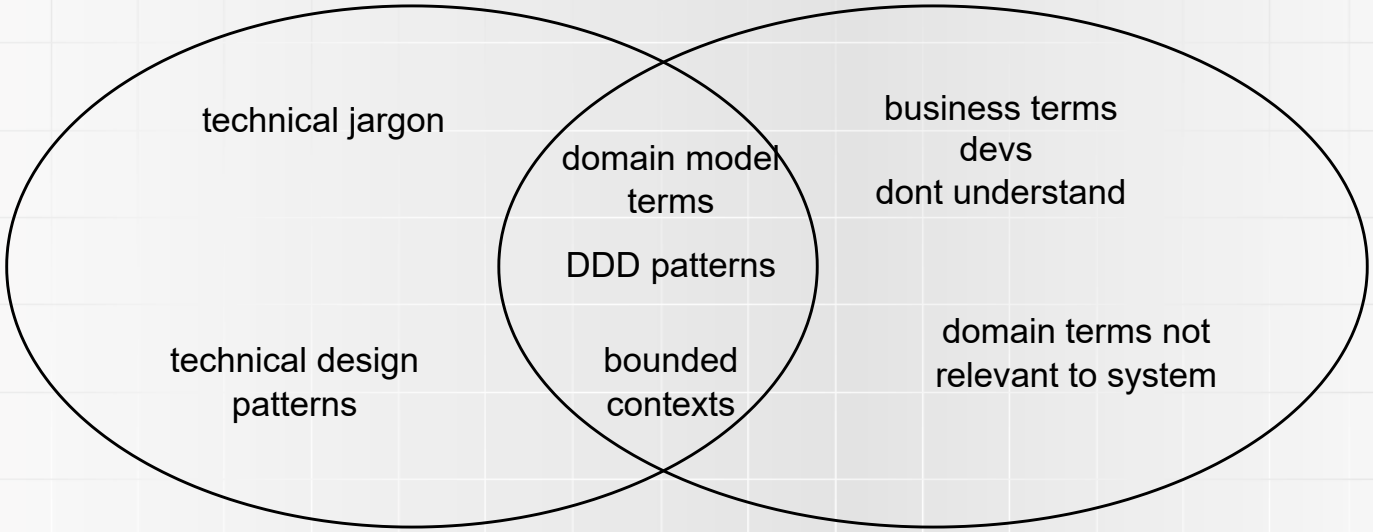


Figure: Ubiquitious Language

DDD Building Blocks

- **Entity**: Defined by identity, not attributes (e.g., User with ID).
- **Value Object**: Defined by attributes, immutable, no identity (e.g., Money, Address).
- **Aggregate**: A consistency boundary that groups entities and value objects and enforces domain invariants.
- **Aggregate Root**: The single entry point to an aggregate; all external access goes through the root.
- **Bounded Context**: Explicit boundary within which a domain model is defined and applicable.

Within a bounded context:

- terms have a **precise and shared meaning**,
- the model is **logically unified**,
- code and data follow the same conceptual rules.

Outside the boundary, **different models may apply** and **explicit translation** is required.

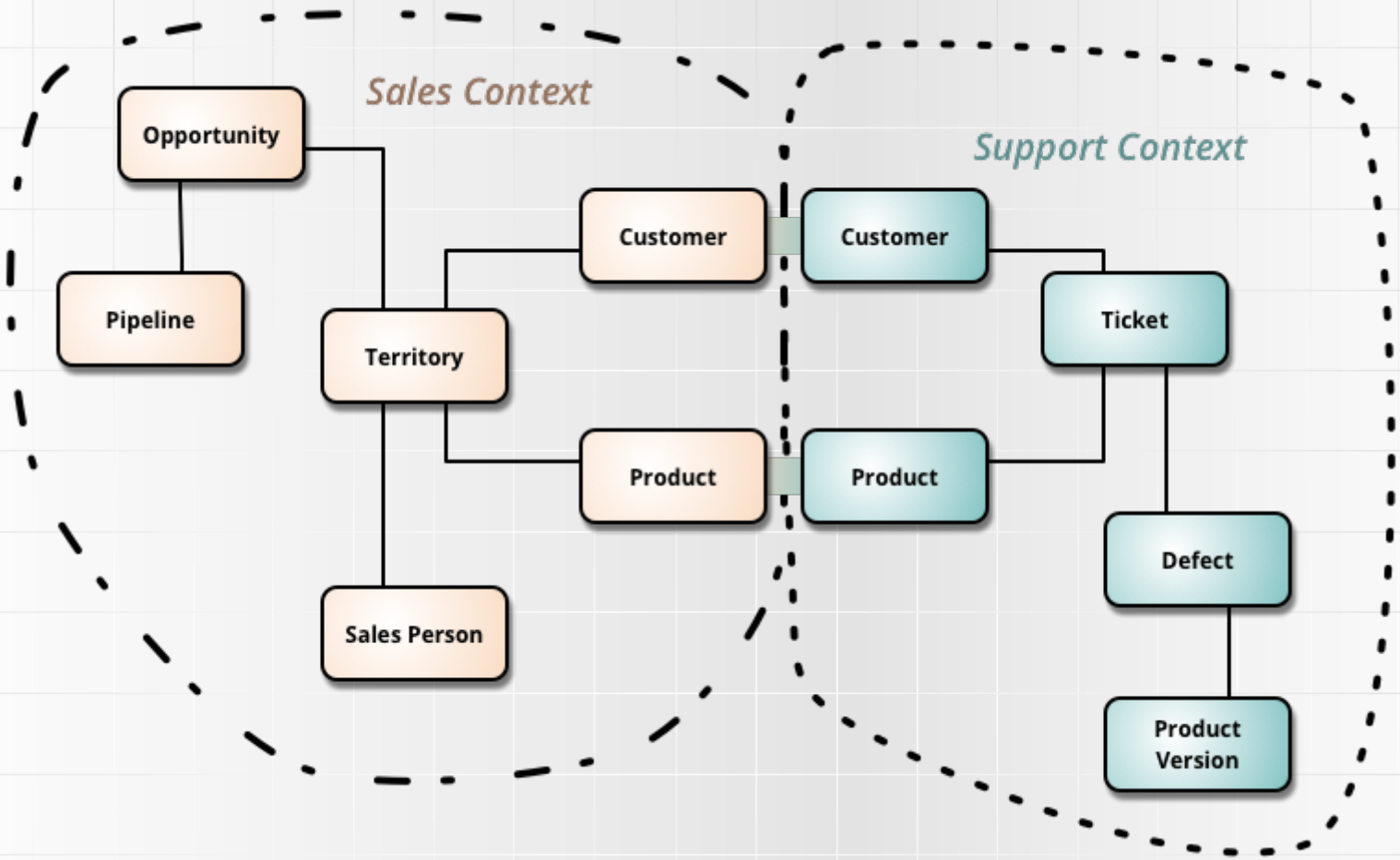


Figure: Source: M. Fowler

- Notice **Product** and **Customer** – appear in both contexts because they represent different domain concepts with different meanings, even though they share the same name.

Domain-Driven Design

Strategic vs Tactical

Aspect	Strategic DDD	Tactical DDD
Primary focus	System structure and boundaries	Domain model and code structure
Main question	*Where does a model apply?*	*How is the model implemented?*
Key concepts	Bounded Context, Context Map	Entity, Value Object, **Aggregate**, Repository
Level	Architecture, teams, integration	Classes, methods, invariants
Typical decisions	Context boundaries, integration style	Encapsulation, consistency rules
Failure mode	Conceptual chaos, mixed models	Anemic domain model

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Modern Architectural Approaches

Modern Architectural Approaches

Clean Architecture

Clean Architecture is an architectural approach that aims to **separate business logic from technical concerns** and **protect the core domain from external dependencies**.

Key principles

- Dependencies always point **inward**
- Business rules are **independent of frameworks**
- The core domain is **testable in isolation**

Typical layers

- **Entities** — enterprise business rules
- **Use Cases** — application-specific rules
- **Interface Adapters** — controllers, presenters
- **Frameworks & Drivers** — UI, database, external systems

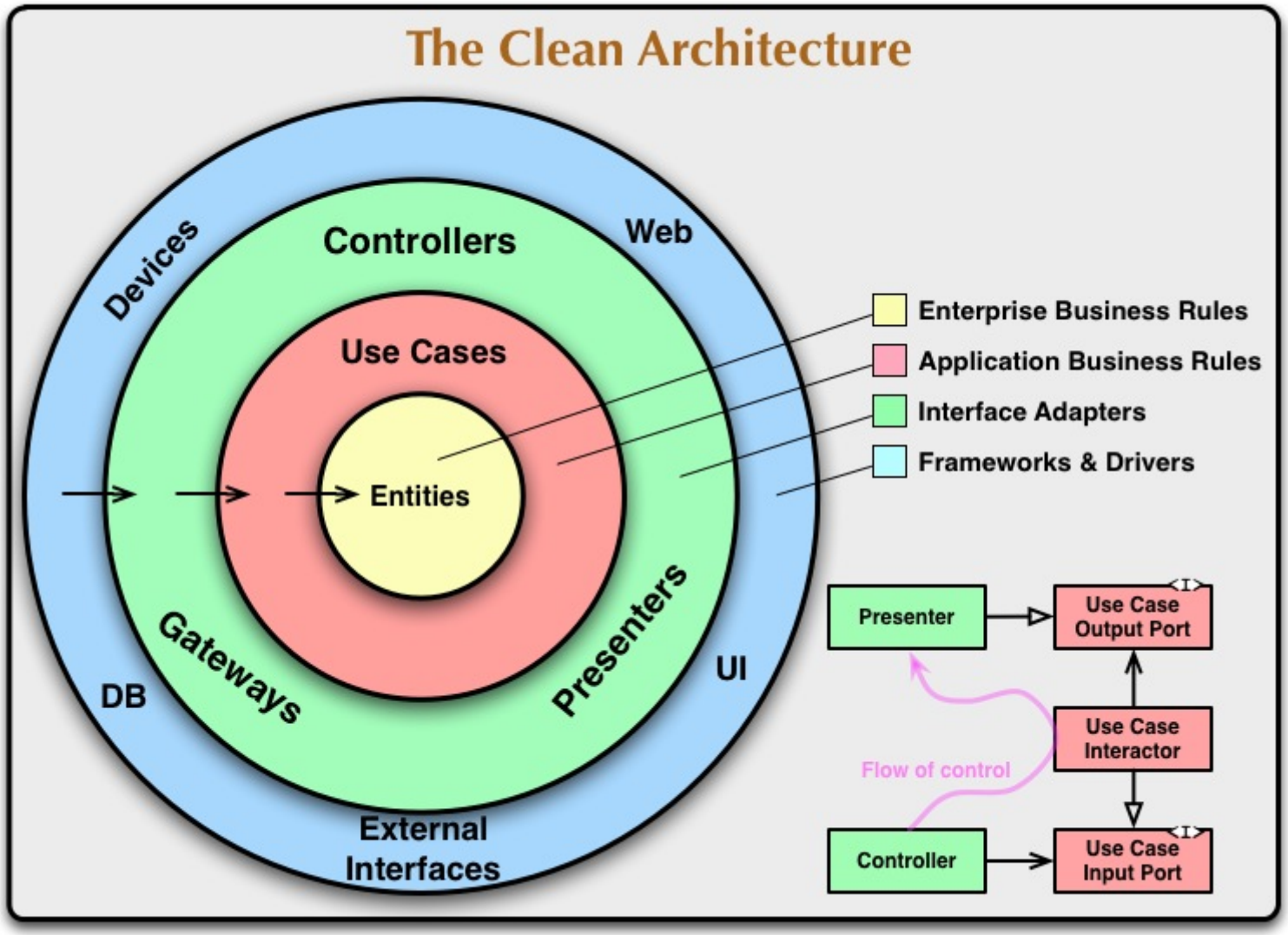

Figure: Source: R. Martin

Command Query Responsibility Segregation (CQRS)

CQRS is an architectural pattern that separates **write operations (commands)** from **read operations (queries)**.

Motivation

- Different requirements for **writing** and **reading** data.
- Complex domain logic on writes, performance-oriented reads.
- Avoiding overly complex and overloaded domain models.

Core idea

- **Commands** change system state (no return value).
- **Queries** return data (no side effects).
- Separate models for reads and writes.

Consequences

- ✓ Clear separation of responsibilities.
- ✓ Better scalability and performance tuning.
- × Increased complexity and duplication.
- × Harder consistency management.

CQRS

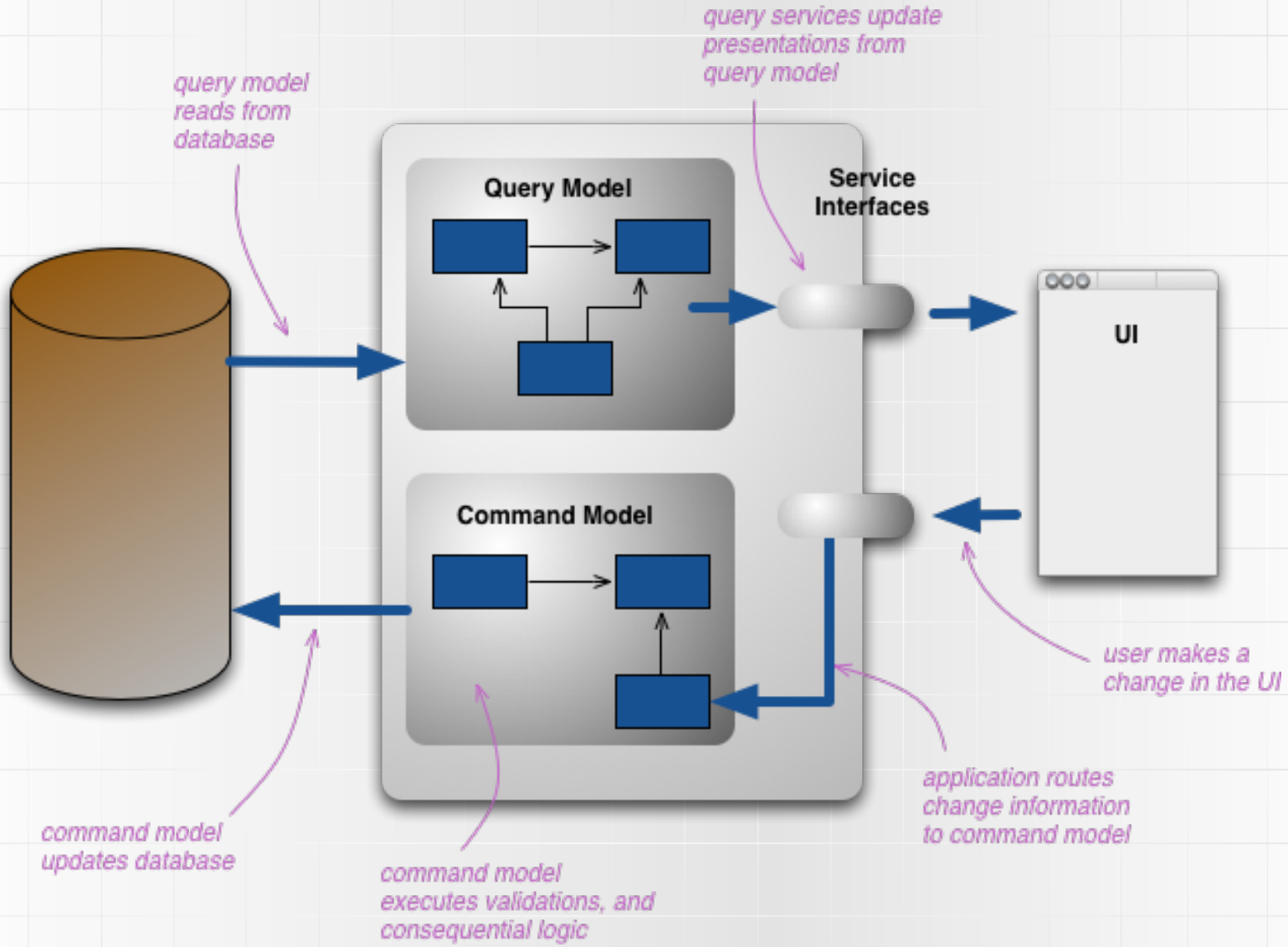

Figure: Source: M Fowler

Dependency Injection (DI)

Dependency Injection: definition

A design technique in which an object *receives* its dependencies instead of creating them.

Core idea: *Inversion of Control* – separation of **object construction** from **object behavior**.

- Dependencies are expressed as **interfaces**, not concrete implementations.
- Improves **testability** (substitution, mocking, isolation).
- Enables **loose coupling** and modular architecture.
- **Frameworks** (e.g. Spring, .NET, Angular) often manage object creation and wiring via *containers*.

Dependency Injection

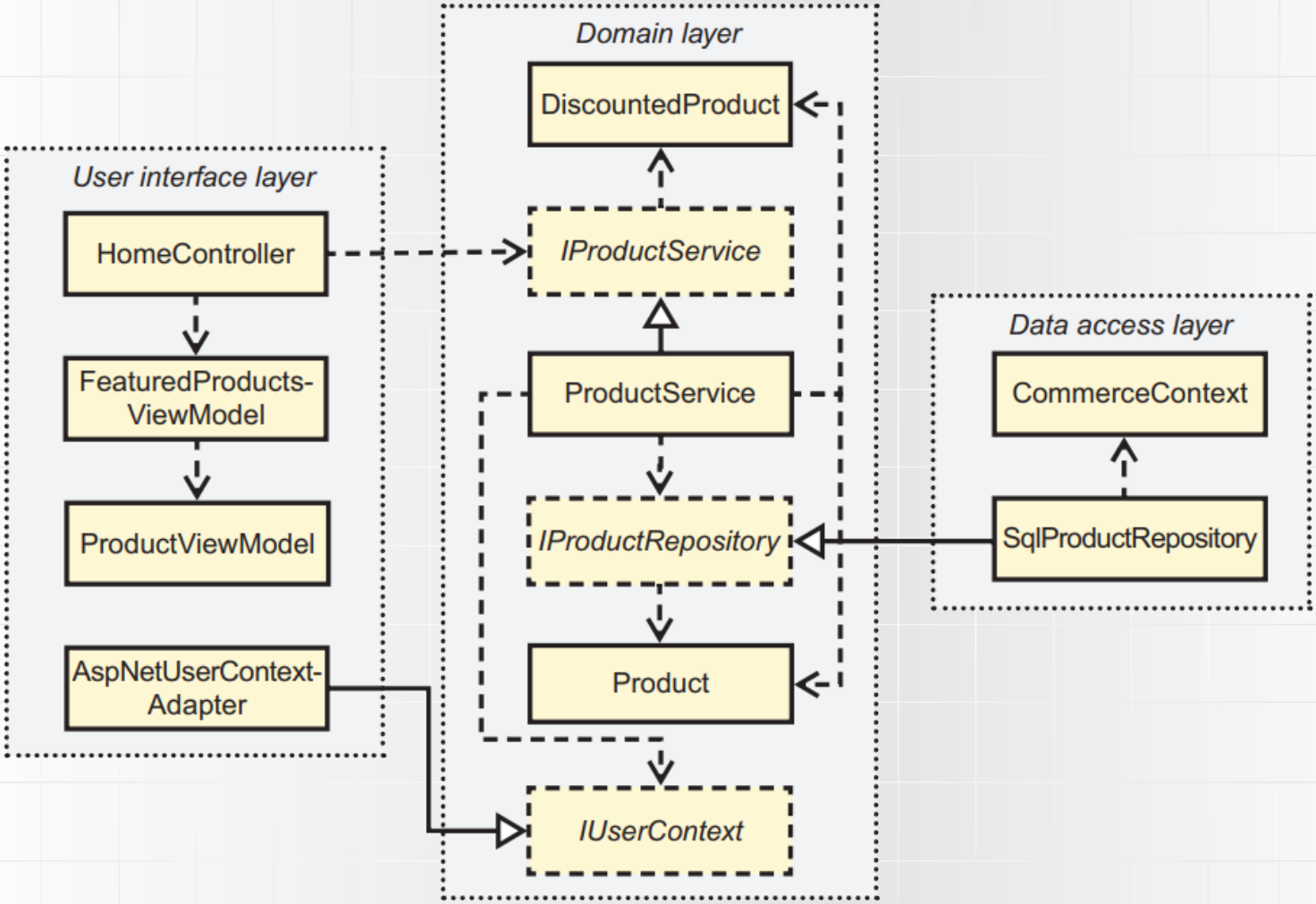

Figure: Source: Mark Seemann, Steven van Deursen - Dependency Injection Principles, Practices, and Patterns, Manning Publications, 2019

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It's all for today!